

Got it — I'll create a **Standard Operating Procedure (SOP)** for a DevOps CI/CD pipeline. I'll make it generic enough so it can apply whether you're using **Jenkins, GitHub Actions, GitLab CI, or Azure DevOps**, but still detailed enough for practical use.

Standard Operating Procedure (SOP)

Title: CI/CD Pipeline Management

Version: 1.0

Effective Date: [Insert Date]

Prepared By: [Your Name / Team]

Approved By: [Manager / Lead]

1. Purpose

To define the standardized process for building, testing, and deploying software using the CI/CD pipeline, ensuring consistent, reliable, and automated delivery of code to production.

2. Scope

This SOP applies to all software development projects maintained by the DevOps team. It covers the **source code integration, build process, automated testing, deployment, and post-deployment validation**.

3. Responsibilities

Role	Responsibility
Developer	Commit and push code changes following branch strategy; write unit tests; ensure code passes local checks before push.
DevOps Engineer	Maintain and optimize CI/CD pipeline; manage environment configuration; ensure deployment automation.
QA Engineer	Validate automated test results; conduct manual testing where needed.
Release Manager	Approve production deployments; ensure rollback plans are in place.

4. Tools & Technologies

- Source Control:** GitHub / GitLab / Bitbucket
- CI/CD Orchestrator:** Jenkins / GitHub Actions / GitLab CI / Azure DevOps
- Build Tools:** Maven / Gradle / npm / pip

- **Testing Frameworks:** JUnit / PyTest / Jest / Selenium
 - **Deployment:** Kubernetes / PCF / AWS EC2 / NGINX buildpack
 - **Monitoring:** Prometheus / Grafana / New Relic
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5. Procedure

5.1 Code Commit & Merge

1. Create a feature branch from develop or main branch.
 2. Follow **commit message guidelines** (e.g., Conventional Commits).
 3. Push changes to remote repository.
 4. Create a **Pull Request (PR)** or **Merge Request (MR)** for code review.
 5. At least **one peer review** is required before merging.
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5.2 CI Build Process

1. CI pipeline triggers automatically on PR or commit to main branches.
 2. Steps executed:
 - **Checkout code** from repository.
 - **Install dependencies** (e.g., npm install, mvn install).
 - **Run unit tests**; fail pipeline if coverage < 80%.
 - **Build artifact** (e.g., .jar, .war, Docker image, zip).
 - **Store artifact** in artifact repository (e.g., Nexus, JFrog Artifactory, S3).
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5.3 Automated Testing

1. Run static code analysis (e.g., SonarQube).
 2. Run integration tests.
 3. Run security scans (e.g., OWASP Dependency Check, Snyk).
 4. Generate and store test reports.
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5.4 Deployment

Staging Deployment

1. Deploy to **staging** environment automatically after successful build and tests.
2. Run smoke tests to verify deployment.

3. Notify QA team for functional testing.

Production Deployment

1. Deployment to **production** requires approval from **Release Manager**.
 2. Deploy artifacts using infrastructure automation (e.g., Ansible, Helm charts, PCF CLI).
 3. Verify deployment via post-deployment health checks.
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5.5 Rollback Procedure

1. If deployment fails, revert to the last stable version from artifact repository.
 2. Notify stakeholders of rollback.
 3. Perform root cause analysis before next deployment.
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6. Quality Control & Monitoring

- **Pipeline Failure Alerts:** Email/Slack notifications.
 - **Log Monitoring:** Check logs for build/test/deploy errors.
 - **Performance Monitoring:** Ensure no regression post-deployment.
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7. Change Management

- All pipeline changes must be tested in a **sandbox environment** before applying to production.
 - Major changes require approval from **DevOps Lead**.
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8. References

- Git Branching Strategy Document
 - Test Automation Guidelines
 - Release Management Policy
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9. Revision History

Version	Date	Author	Description
1.0	[Date]	[Your Name]	Initial SOP draft

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Do you want me to prepare it in a **downloadable DOCX format** so it's ready to share with your DevOps team?